

Discovery Executive Summary

Project: updated-discovery · Generated: 18/08/2026, 14:08:26

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

Portfolio Overview

#	Analysis	Overall Rating
1	Architecture & Design Analysis	—

1. Architecture & Design Analysis

EXECUTIVE SUMMARY

The Klearcom platform is a monorepo with three layers: a Laravel 12 PHP backend (15 PHP files), a React 19 TypeScript frontend (15 TS/TSX/JSX files), and a Node.js Express development API (6 JS files). While the controllers are lean in line count (avg 63 LOC) and basic dependency injection is present via Laravel constructor injection, two structural risks dominate: **zero repository classes** exist, scattering 35 direct ORM access points across controllers and services, and **3 of 7 data stores (43%) are shared** across the Discovery and Connect domains via multi-module MongoDB collections with no ownership boundaries. The `app/Modules/Connect/` and `app/Modules/Discovery/` directories exist but contain no code — an abandoned domain-driven structure that leaves all models, controllers, and services in a flat shared namespace. On the frontend, page components embed all data-fetching, mutation orchestration, and cache invalidation inline with no service or data-access layer. Duplicated business logic (reachability calculation formula, `buildTree` function) appears in three independent locations each, creating divergence risk.

1.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	Good	Moderate	High Risk	Measured	Rating
H1	Fat Controllers	Avg LOC per controller	<150	150–300	>300	63 LOC	Good
H2	Missing Service Layer	Controllers accessing repos/models	<15	15–30	>30	25	Moderate
H3	Missing Repository Pattern	Direct DB access points outside repos	<15	15–30	>30	35	High Risk
H4	Circular Dependencies	Dependency cycles	0	1–3	>3	0	Good
H5	Shared Utility Abuse	Utility files w/ business logic	0	1–5	>5	1	Moderate
H6	Direct SQL in Controllers	ORM compliance %	>90%	60–90%	<60%	100%	Good
H7	God Classes	Largest class/file LOC	<500	500–600	>600	276 LOC	Good
H8	Domain Boundary Violations	Cross-domain access points	0	1–5	>5	5	Moderate
H9	Shared Database Coupling	Tables shared across domains	<30%	30–40%	>40%	43% (3/7)	High Risk
H10	Duplicated Business Logic (additional)	Distinct business logic fragments duplicated ≥2×	0	1–3	>3	3	Moderate

No additional hotspots beyond H10 were observed.

1.4 Actions Required

Hotspot	Action	Rating	Priority
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H3. Missing Repository Pattern	Create <code>ConnectMonitorRepository</code> , <code>ConnectCheckResultRepository</code> , <code>DiscoveryJobRepository</code> , <code>DiscoveryNodeRepository</code> with interfaces; refactor all 35 direct ORM access points to use repositories	High Risk	Critical
H9. Shared Database Coupling	Split 3 shared MongoDB collections into domain-owned collections or add domain-specific ACL wrapper services; define ownership per collection	High Risk	Critical
H2. Missing Service Layer	Create <code>ConnectService</code> , <code>DiscoveryService</code> , <code>DashboardService</code> application services; move 25 controller model calls through services; introduce frontend service modules for 4 page components	Moderate	High
H10. Duplicated Business Logic	Extract <code>ReachabilityCalculator</code> (3 copies), consolidate <code>buildTree</code> (3 copies), create shared frontend hook pattern (2 copies)	Moderate	High
H8. Domain Boundary Violations	Populate <code>app/Modules/Connect/</code> and <code>app/Modules/Discovery/</code> with domain models, services, and repositories; define published interfaces for cross-domain reads	Moderate	Medium
H5. Shared Utility Abuse	Replace <code>extract()</code> in <code>LegacyDataMapper</code> and <code>LegacyReportController</code> with explicit destructuring; move report mapping into a <code>ReportingService</code>	Moderate	Medium

1.5 Expected Outcomes

- **Testability:** Repository interfaces enable unit testing services without hitting the database; constructor DI (already partially in place) becomes complete.
- **Domain independence:** Populated `app/Modules/` with owned models, services, and repositories allows Discovery and Connect to evolve independently — schema changes in one domain cannot silently break the other.
- **Single source of truth for business logic:** A canonical `ReachabilityCalculator` eliminates 3 divergent copies of the reachability formula; `buildTree` lives in one place.
- **Frontend maintainability:** Domain service modules centralize query keys, cache invalidation, and API orchestration — page components shrink to thin UI views.
- **Microservice readiness:** Bounded contexts with published interfaces and anti-corruption layers make it possible to extract Discovery or Connect into standalone services when scale demands it.

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